Bearing temperature drift on the HX-400 pump series

Technical Note TN-2026-014 · Halvorsen Fluid Systems, Reliability Engineering

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Applies to	HX-400, HX-420 and HX-460 centrifugal pumps
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1 Summary

Since the autumn of 2025, field engineers have reported a slow rise in drive-end bearing temperature on HX-400 series pumps that run continuously at high duty. The rise is gradual, typically eight to twelve degrees over three to five months, and in most cases it stays below the trip limit. In a small number of installations, all of them HX-420 units in hot plant rooms, it ended in a bearing seizure.

The investigation traced the drift to the shielded drive-end bearing fitted since mid-2024, which loses its grease charge faster than the sealed bearing it replaced when the ambient temperature is high. This note gives revised alarm thresholds for all three models, introduces the sealed bearing BRG-6207-2RS as the standard drive-end part, and shortens the re-greasing interval for the HX-420.

2 Background

2.1 The HX-400 series

The HX-400 series is a family of end-suction centrifugal pumps for process water, brewing and food applications. The three models share the same bearing frame and differ in impeller diameter and motor size. The HX-400 is the base model, the HX-420 has a larger impeller for higher head, and the HX-460 is a high-flow variant with a reinforced shaft. All three use a deep-groove ball bearing at the drive end and an angular-contact pair at the non-drive end.

2.2 How bearing temperature is monitored

Every pump shipped since 2023 carries a temperature probe in the drive-end bearing housing. The probe reports to the pump controller, which raises a warning when the temperature crosses the warning threshold and stops the pump when it crosses the trip threshold. Sites with a Halvorsen telemetry gateway also send the readings to the remote monitoring service, which keeps a twelve-month history for each pump. That history is what made the slow drift visible: a single reading never looked abnormal, but the trend over several months did.

3 Findings

3.1 Field observations

Reliability Engineering reviewed the monitoring history of 214 pumps across 38 sites. Pumps fitted with the shielded bearing BRG-6207-ZZ showed a steady upward trend in drive-end temperature once they passed roughly 1,500 operating hours, while pumps with the earlier sealed bearing stayed flat. The effect was strongest on HX-420 units, which run the hottest of the three because of their higher shaft power, and on sites where the plant room stays above 35 °C in summer.

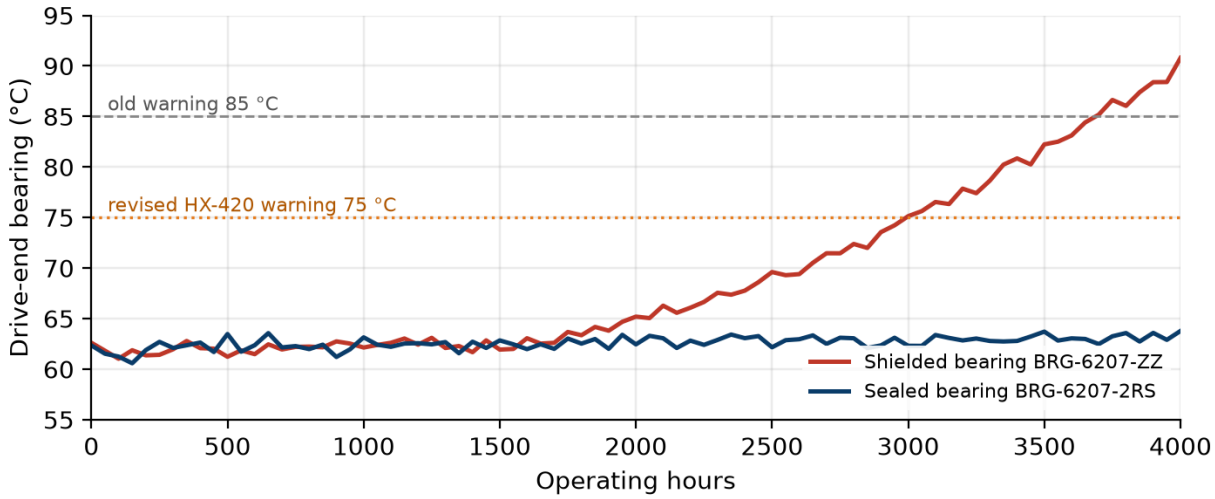


Figure 1: Drive-end bearing temperature of two HX-420 pumps on the same site, one with the shielded bearing BRG-6207-ZZ and one with the sealed bearing BRG-6207-2RS, over the first 4,000 operating hours.

Figure 1 shows the pattern seen across the affected sites. Both pumps start at the same temperature. After about 1,500 hours the pump with the shielded bearing begins to drift upwards and crosses the old warning threshold shortly after 3,000 hours, while the pump with the sealed bearing stays within two degrees of its starting value.

3.2 Root cause

Teardown of six failed bearings showed dry, darkened grease and polished raceways, the typical signature of lubricant starvation rather than overload or misalignment. The shielded bearing has a small gap between the shield and the inner ring. At high operating temperature the base oil of the grease migrates out through that gap, and the remaining thickener no longer lubricates the rolling elements. Friction then rises, which raises the temperature further and speeds up the loss of oil.

The sealed bearing has a contact seal instead of a gap and keeps its grease charge for the full service interval.

Shaft alignment, impeller balance and motor condition were checked on every failed unit and were within specification. The bearing change in mid-2024 was a supply decision and was not tested at high ambient temperature, which is why the problem only appeared after a full summer of operation.

4 Revised alarm thresholds

The original thresholds were set for the sealed bearing and left too little margin to catch a drift before it became a failure. The revised values below apply to all pumps in the series, whatever bearing they carry, and take effect with the controller parameter set released alongside this note.

Table 1: Drive-end bearing temperature thresholds, old and revised

Model	Warning		Trip		Margin before trip, revised
	Old	Revised	Old	Revised	
HX-400	85 °C	80 °C	95 °C	90 °C	10 K
HX-420	85 °C	75 °C	95 °C	88 °C	13 K
HX-460	90 °C	82 °C	100 °C	92 °C	10 K

The HX-420 gets the lowest warning threshold because it is the model where the drift starts earliest. A warning on its own does not require the pump to be stopped. It does require a bearing

inspection at the next planned stop, and the engineer should check the trend in the monitoring history rather than the single reading.

5 Corrective action

5.1 Replacement bearing

The sealed bearing BRG-6207-2RS is now the standard drive-end bearing for the whole series. It has been fitted in production from serial number HX4-26-0410 onwards. Pumps below that serial number keep the shielded bearing BRG-6207-ZZ until their next bearing change, at which point the sealed bearing must be fitted instead. The two parts are dimensionally identical and no other component changes, so the replacement can be done during a routine service visit.

5.2 Re-greasing interval

Until the shielded bearing has been replaced, the drive-end bearing must be re-greased more often than the maintenance manual specifies. Once the sealed bearing is fitted, the manual's interval applies again for the HX-400 and HX-460, but the HX-420 keeps a shorter interval because of its higher running temperature.

Table 2: Drive-end re-greasing interval and grease quantity

Model	Interval in operating hours			Grease per re-greasing	
	Manual	Shielded bearing fitted	Sealed bearing fitted	Shielded	Sealed
HX-400	4,000 h	2,000 h	4,000 h	8 g	12 g
HX-420	4,000 h	1,500 h	2,500 h	8 g	12 g
HX-460	3,000 h	1,500 h	3,000 h	10 g	15 g

Use the grease specified in the maintenance manual, in the smaller quantity while the shielded bearing is fitted: a fuller charge only pushes more grease past the shield. Mixing greases with different thickeners can make the problem worse, so if the type in the bearing is unknown, clean it out completely before re-greasing.

6 Affected serial ranges

The shielded bearing was fitted in production between July 2024 and February 2026. The affected ranges are:

- HX-400: serial numbers HX4-24-0612 to HX4-26-0409
- HX-420: serial numbers HX4-24-0655 to HX4-26-0398
- HX-460: serial numbers HX4-24-0701 to HX4-26-0402

Bearings supplied as spare parts in the same period may also be the shielded type. Check the part number on the bearing itself rather than relying on the service record.

7 Revision history

Revision A, 12 March 2026: first issue.